Master Project - RMP LoAI

Principal Component Analysis (PCA) and Cluster Analysis of Africa World Development Indicators (WDI) Dataset

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1 Introduction

This project demonstrates Principal Component Analysis (PCA) and cluster analysis of the Africa World Development Indicators (WDI) dataset. PCA is a statistical technique used to reduce many correlated variables into a smaller number of components while retaining as much of the original information/variation as possible. The WDI dataset has more than 30 development indicators. PCA can help answer: "Can these indicators be summarized into a smaller number of underlying dimensions of development?" For example, the variables might reduce to components roughly representing: Economic development, Health and human development, Education, Infrastructure/access. Then we can use those components for further analysis, including cluster analysis. Cluster analysis is a statistical technique used to group observations into relatively homogeneous groups (clusters) based on their similarity across several variables. In this project, after the PCA, we will group countries that have similar development characteristics. For example, after the analysis, we might find groups such as countries with relatively similar economic, health, education and infrastructure profiles.

Table 1: Missing Value Analysis

Variable	Missing_n	Missing_percent
country	0	0
country_code	0	0
GDP_growth	2	3.704
GDP_percapita	2	3.704
oil_rents	3	5.556
oda	1	1.852
proportion_female	1	1.852
life_expectancy	0	0
education_expenditure	13	24.074
primary_school_completion	27	50
pupilteacher_ratio	26	48.148
literacy	26	48.148
electricity	0	0
trade	5	9.259
military_expenditure	6	11.111
CO2_percapita	0	0
multidimensional_poverty	52	96.296
physicians	33	61.111
tetanus	1	1.852
prenatalcare	36	66.667
HIV_15_49	1	1.852
antiretroviral_HIV	2	3.704
suicide	0	0
food_insecurity	24	44.444
urbanpop	1	1.852
urbanpop_growth	1	1.852
ruralpop	1	1.852
pop_growth	1	1.852
population	1	1.852
subsaharan	0	0
hdi	1	1.852
freedomhouse	0	0
basicfreedom	0	0

After missing value analysis, variables with more than 20% missing values were excluded. From Tables 2 and 3 below, Country-level missing-value analysis showed that missingness was present in most countries, but was generally limited in extent. The median country had four missing indicator observations out of the 31 variables included in the analysis, while the mean was 4.94. Most countries had between three and six missing observations.

Only one country had complete information across all 31 indicators, while Er-

itrea had the highest level of missingness, with 16 missing observations (51.6%). Because complete-case analysis would exclude 53 of the 54 countries, it was not considered appropriate for this analysis.

Table 2: Missing Value Summary

Statistic	N	Mean	St. Dev.	Min	Max
Missing_n	54	4.944	2.666	0	16
Missing_percent	54	15.950	8.602	0.000	51.613

Table 3: Missing Values by Country

Country	Missing_n	Missing_percent
Algeria	2	6.452
Angola	6	19.355
Benin	3	9.677
Botswana	5	16.129
Burkina Faso	1	3.226
Burundi	5	16.129
Cabo Verde	3	9.677
Cameroon	3	9.677
Chad	6	19.355
Central African Republic	6	19.355
Comoros	7	22.581
Congo	5	16.129
Democratic Republic of the Congo	7	22.581
Cote d'Ivoire	4	12.903
Djibouti	6	19.355
Eritrea	16	51.613
Equatorial Guinea	8	25.806
Eswatini	4	12.903
Ethiopia	5	16.129
Gabon	6	19.355
Gambia	3	9.677
Ghana	0	0
Guinea	4	12.903
Guinea-Bissau	7	22.581
Kenya	4	12.903
Lesotho	6	19.355
Liberia	8	25.806
Madagascar	3	9.677
Mali	3	9.677
Malawi	5	16.129
Mauritius	2	6.452
Mauritania	4	12.903
Morocco	3	9.677
Mozambique	2	6.452
Namibia	3	9.677
Niger	6	19.355
Nigeria	4	12.903
Rwanda	3	9.677
Sao Tome and Principe	9	29.032
Senegal	4	12.903
Sierra Leone	2	6.452
Seychelles	5	16.129
Somalia	11	35.484
South Sudan	10	32.258
South Africa	5	16.129
Sudan	5	16.129
Tanzania	4	12.903
Togo	4	12.903
Tunisia	5	16.129
Uganda	4	12.903
Zambia	4	12.903
Zimbabwe	4	12.903
Egypt	4	12.903
Libya	9	29.032

Scale Assessment

The descriptive statistics below showed substantial differences in the measurement scales and variability of the 31 indicators. For example, population and GDP per capita exhibited values and standard deviations several orders of magnitude larger than indicators such as HDI and basic freedom. Because PCA is sensitive to the scale of the variables, using the indicators in their original units could allow variables with larger numerical variance to exert disproportionate influence on the extracted components. The indicators were therefore standardized before conducting PCA.

Table 4: Scale assessment

Variable	Minimum	Maximum	Mean	SD
GDP_growth	-6.237	15.133	3.822	3.435
GDP_percapita	862, 942, 909.000	1, 150, 000, 000, 000	120, 354, 457, 476.000	235, 815, 635, 882.000
oil_rents	0	42.780	4.119	9.870
oda	6, 720, 000.000	4, 941, 029, 785	909, 782, 591.000	940, 459, 795.000
proportion_female	44.456	52.357	50.117	1.129
life_expectancy	52.805	76.693	63.763	5.983
education_expenditure	1.566	8.988	4.118	1.448
primary_school_completion	40.560	105.144	80.672	11.962
pupilteacher_ratio	14.488	59.509	35.521	8.749
literacy	34.523	95.868	71.968	15.577
electricity	6.173	100	53.360	26.494
trade	1.295	288.217	74.776	42.938
military_expenditure	0.162	5.468	1.586	0.912
CO2_percapita	0.026	8.825	1.211	1.909
multidimensional_poverty	31.990	47.800	39.895	1.536
physicians	0.036	2.533	0.291	0.502
tetanus	60	100	85.963	8.823
prenatalcare	67	99	89.544	5.017
HIV_15_49	0.100	28	4.104	6.282
antiretroviral_HIV	10	94	54.926	20.832
suicide	1.500	76.600	9.017	10.544
food_insecurity	17.600	88.200	52.752	14.477
urbanpop	13.032	89.370	45.925	18.651
urbanpop_growth	-0.063	6.186	3.396	1.206
ruralpop	10.630	86.968	54.075	18.651
pop_growth	0.055	3.817	2.306	0.818
population	96, 762	195, 874, 685	23, 762, 311.000	33, 954, 068.000
subsaharan	0	1	0.926	0.264
hdi	0.391	0.801	0.557	0.105
freedomhouse	2	90	43.148	25.193
basicfreedom	0	1	0.426	0.499

Standardization of Variables

Table 5 below shows the standardized variables. After standardization, each variable has a mean of approximately 0 and a standard deviation of 1. This puts all 31 indicators on the same scale, regardless of their original units or magnitude.

Therefore, no variable will dominate the PCA simply because it has larger numerical values or greater variance due to its measurement scale. The PCA can instead identify patterns based on the standardized variation and relationships among the indicators.

Table 5: New Scale

Variable	Mean	SD
GDP_growth	0	1
GDP_percapita	0	1
oil_rents	0	1
oda	-0	1
proportion_female	0	1
life_expectancy	0	1
education_expenditure	-0	1
primary_school_completion	-0	1
pupilteacher_ratio	0	1
literacy	0	1
electricity	0	1
trade	0	1
military_expenditure	0	1
CO2_percapita	0	1
multidimensional_poverty	0	1
physicians	0	1
tetanus	0	1
prenatalcare	-0	1
HIV_15_49	-0	1
antiretroviral_HIV	0	1
suicide	0	1
food_insecurity	-0	1
urbanpop	0	1
urbanpop_growth	0	1
ruralpop	-0	1
pop_growth	0	1
population	-0	1
subsaharan	-0	1
hdi	0	1
freedomhouse	0	1
basicfreedom	0	1

Assessment of Population-Structure Variables
A correlation analysis was conducted to examine the relationships among the standardized development indicators. The analysis revealed a perfect negative correlation between *urbanpop* and *ruralpop* ($r = -1.000$), indicating that the two variables provide essentially complementary information about population distribution. Because retaining both variables would introduce perfect redundancy into the PCA input, *ruralpop* was excluded from the PCA dataset, while *urbanpop* was retained. The exclusion was applied only to the analytical dataset used for PCA and did not alter the original dataset. Following this adjustment, the PCA was conducted using 30 indicators.

PCA Adequacy Diagnostics
The KMO indicates only marginal adequacy, suggesting that the common structure among the indicators is not particularly strong. However, Bartlett's test is highly significant, demonstrating that the correlation matrix contains substantial relationships and is not an identity matrix. Taken together, these results provide sufficient justification to proceed with PCA as an exploratory analysis, while recognizing that the relatively low KMO warrants caution when interpreting the resulting components.

Table 6: Kaiser-Meyer-Olkin (KMO) Measure

Measure	Value
Overall KMO	0.529

Table 7: Bartlett's Test of Sphericity

Test	Chi_square	df	p_value
Bartlett's test of sphericity	1180.442	435	< 0.001

PCA
Principal Component Analysis (PCA) was conducted on the 30 standardized development indicators. The analysis was used to identify a smaller number of dimensions that could summarize the variation contained in the original indicators.

Table 8 below shows that The PCA identified nine components with eigenvalues greater than 1, ranging from 7.706 for PC1 to 1.135 for PC9. PC1 accounted for the largest proportion of variance (24.86%), followed by PC2 (12.01%) and PC3 (9.94%). The first four components accounted for 54.05% of the total variance, while the nine components with eigenvalues greater than 1 collectively explained 78.97%. Although the eigenvalue-greater-than-one criterion therefore

suggests nine potential components, this criterion alone does not determine the final number of components to retain. The scree plot was subsequently examined to assess where the major decline in explained variance levels off.

Table 8: PCA results

Component	Eigenvalue	Variance	Cumulative
PC1	7.706	24.860	24.860
PC2	3.725	12.010	36.870
PC3	3.081	9.940	46.810
PC4	2.245	7.240	54.050
PC5	1.975	6.370	60.430
PC6	1.890	6.100	66.520
PC7	1.440	4.650	71.170
PC8	1.284	4.140	75.310
PC9	1.135	3.660	78.970

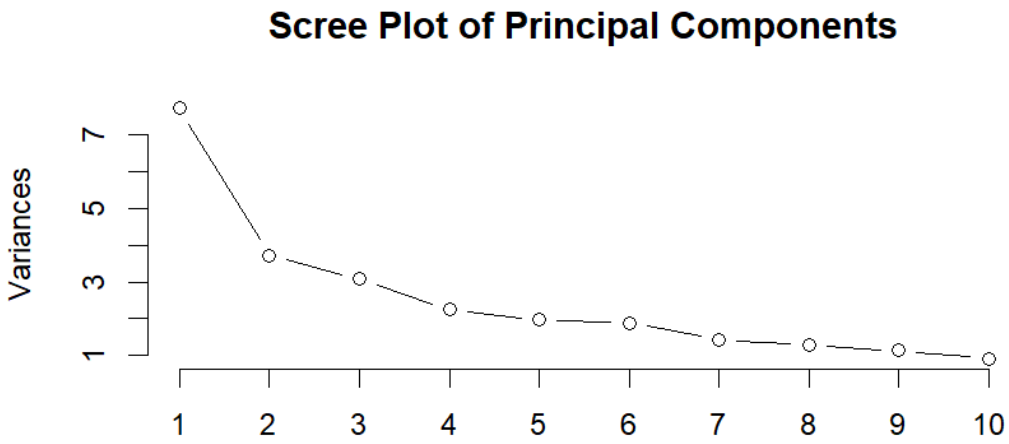


Figure 1: Scree Plot of Principal Components

The scree plot shows a large decline in eigenvalues from PC1 through the first few components, followed by a much more gradual decline. The curve begins to flatten around PC4, indicating that the first four components capture the main

structure in the data, while subsequent components contribute progressively smaller amounts of additional variance.

Table 9 below shows the loading structure shows that the four retained components represent different dimensions of country-level development. PC1 is mainly associated with broad development and living conditions, with strong contributions from HDI, electricity access, life expectancy, and CO per capita. PC2 reflects a more mixed combination of health, social, educational, and institutional characteristics, particularly HIV prevalence, suicide, education expenditure, and freedom-related indicators. PC3 is primarily associated with economic scale and development financing, driven by population, ODA, and GDP per capita. PC4 combines health and socioeconomic characteristics, with stronger contributions from HIV prevalence, suicide, GDP growth, tetanus coverage, and life expectancy.

Table 9: PCA Results by indicator

Indicator	PC1	PC2	PC3	PC4
GDP_growth	0.033	-0.015	0.159	-0.325
GDP_percapita	-0.080	-0.237	0.372	0.220
oil_rents	-0.065	-0.270	-0.229	0.208
oda	0.143	-0.130	0.392	0.002
proportion_female	0.062	0.259	0.177	-0.033
life_expectancy	-0.265	-0.069	0.094	-0.279
education_expenditure	-0.130	0.270	0.170	0.155
primary_school_completion	-0.217	0.005	0.168	0.094
pupilteacher_ratio	0.247	0.060	0.002	0.034
literacy	-0.185	0.077	-0.009	-0.032
electricity	-0.311	-0.066	0.076	-0.029
trade	-0.150	0.097	-0.268	0.042
military_expenditure	-0.052	-0.045	-0.082	0.188
CO2_percapita	-0.265	-0.080	-0.005	0.179
multidimensional_poverty	0.075	-0.025	0.018	-0.094
physicians	-0.231	-0.019	0.114	0.019
tetanus	-0.173	0.187	0.051	-0.318
prenatalcare	-0.096	0.176	-0.178	-0.228
HIV_15_49	-0.035	0.337	0.033	0.391
antiretroviral_HIV	-0.030	0.199	0.195	0.072
suicide	-0.041	0.314	-0.029	0.370
food_insecurity	0.209	0.128	-0.135	0.169
urbanpop	-0.237	-0.173	-0.210	-0.010
urbanpop_growth	0.252	-0.074	0.057	-0.100
ruralpop	0.237	0.173	0.210	0.010
pop_growth	0.232	-0.164	-0.031	-0.135
population	0.059	-0.234	0.428	0.148
subsaharan	0.169	0.245	-0.121	-0.069
hdi	-0.332	-0.024	0.077	0.025
freedomhouse	-0.133	0.260	0.156	-0.236
basicfreedom	-0.086	0.257	0.150	-0.146

Table 10: Country Scores on the Four Retained Principal Components

country	PC1	PC2	PC3	PC4
Algeria	-6.353	-2.984	1.387	0.997
Angola	0.727	-2.276	-2.257	1.219
Benin	1.224	0.288	0.315	-1.246
Botswana	-2.935	2.889	0.174	0.874
Burkina Faso	2.654	0.586	1.000	-0.904
Burundi	3.406	0.780	0.012	-0.342
Cabo Verde	-4.021	1.953	-0.229	-1.949
Cameroon	1.244	-0.896	-0.503	0.057
Chad	3.691	-0.914	-1.876	0.551
Central African Republic	2.862	-0.659	-1.580	1.483
Comoros	0.036	0.540	-0.101	-1.110
Congo	-0.193	-1.649	-3.900	1.952
Democratic Republic of the Congo	2.445	-1.516	1.331	-0.016
Cote d'Ivoire	0.678	0.116	0.200	-0.695
Djibouti	-2.145	-0.522	-3.634	-1.089
Eritrea	1.044	0.140	-1.052	-0.818
Equatorial Guinea	-0.714	-3.166	-4.060	2.593
Eswatini	-0.818	3.797	0.096	3.383
Ethiopia	2.573	-1.045	3.934	-0.466
Gabon	-2.309	-1.725	-2.465	0.549
Gambia	0.568	-0.348	-1.740	-2.141
Ghana	-1.112	0.029	0.384	-2.246
Guinea	2.511	-0.298	-0.344	0.018
Guinea-Bissau	1.513	0.101	-1.259	-0.160
Kenya	0.896	0.324	2.317	-0.164
Lesotho	-0.776	6.022	-0.449	4.576
Liberia	0.601	0.280	-0.906	-1.256
Madagascar	1.697	-0.435	-0.264	-1.257
Mali	2.436	-1.187	0.155	0.093
Malawi	2.894	2.299	0.938	-0.482
Mauritius	-6.486	1.383	1.180	-1.596
Mauritania	0.377	-1.092	-1.211	-0.800
Morocco	-3.990	-2.038	1.032	0.038
Mozambique	2.602	2.402	0.055	0.110
Namibia	-2.053	3.101	0.578	0.988
Niger	3.483	-0.439	0.485	-0.752
Nigeria	2.036	-4.145	5.360	3.205
Rwanda	1.938	0.856	1.003	-1.238
Sao Tome and Principe	-2.350	0.949	-0.567	-1.987
Senegal	-0.540	1.319	0.726	-2.360
Sierra Leone	1.447	1.672	-0.318	-0.223
Seychelles	-7.060	0.976	-0.191	-0.081
Somalia	1.715	-1.106	-0.745	0.191
South Sudan	3.067	-0.264	-0.998	1.953
South Africa	-3.911	1.391	2.486	2.748
Sudan	1.671	-1.578	-0.295	0.498
Tanzania	2.303	0.251	1.897	-1.187
Togo	0.846	0.310	-0.428	-0.208
Tunisia	-4.700	-0.198	-0.333	-2.060
Uganda	2.798	-0.160	1.134	-1.002
Zambia	0.174	1.658	0.298	-0.538
Zimbabwe	0.174	2.019	0.416	1.156

Cluster 5 profile and Cameroon comparison

Cluster 5 comprised 20 countries, including Cameroon, and represented countries with broadly similar profiles across the four principal-component dimensions used for clustering. The cluster exhibited moderate average levels of human development, with mean life expectancy of 64.9 years, HDI of 0.570, electricity access of 54.9%, literacy of 72.1%, and primary school completion of 82.7%. The cluster also showed considerable variation across economic, health, demographic, and governance-related indicators, indicating that membership should be interpreted as multidimensional similarity rather than equivalence across individual indicators. Compared with the Cluster 5 mean, Cameroon displayed several notable differences. Cameroon had lower primary school completion (64.5% versus 82.7%), trade (40.6 versus 73.1), life expectancy (58.9 versus 64.9 years), education expenditure (3.13% versus 4.59%), and physician availability (0.157 versus 0.416). Conversely, Cameroon recorded higher electricity access (62.2% versus 54.9%), urban population share (56.4% versus 43.6%), literacy (77.1% versus 72.1%), and antiretroviral HIV coverage (56.0% versus 53.0%). Its HDI (0.560) was close to the cluster mean (0.570), despite these differences across individual indicators. Overall, the comparison suggests that Cameroon shares a broad multidimensional profile with the countries in Cluster 5 but retains distinctive characteristics at the indicator level. These differences illustrate the value of combining PCA and clustering: countries may be grouped according to their overall patterns while still exhibiting important differences in specific development indicators. The findings are descriptive and should not be interpreted as evidence of causal relationships.